

Convex Programming Models

1 Introduction

Convex programming (or convex optimization) is a powerful, yet tractable, generalization of linear programming. It is a special case of the more general class of non-linear programming problems:

$$\begin{aligned} \min \quad & f(x_1, x_2, \dots, x_n) \\ & g_i(x_1, \dots, x_n) \leq b_i, \quad i = 1, \dots, m \\ & h_i(x_1, \dots, x_n) = b'_i, \quad i = 1, \dots, m' \end{aligned}$$

where f, g_i are arbitrary nonlinear functions of $x_1, \dots, x_n$. A non-linear program can be written more concisely as:

$$\begin{aligned} \min \quad & f(x) \\ & g_i(x) \leq b_i, \quad i = 1, \dots, m \\ & h_i(x) = b'_i, \quad i = 1, \dots, m' \end{aligned}$$

Although very general (try formulate a binary integer program as a non-linear program), this flexibility comes at a computational price: non-linear programming problems are usually very hard to solve.

To the contrary, we have algorithms and software to solve convex programming problems – and convex programs are general enough to encompass many relevant cases. Thus, being able to recognize and formulate convex programs is an important expertise.

2 Models

We discuss here some problems that can be modeled using non-linear programming. In Section ABVSDFSFE, we will discuss whether they are, or can be reduced to, convex programs.

2.1 Portfolio Optimization

Consider the following problem. An investor has \$5000 and two potential investments. From historical data, investments 1 and 2 have an expected annual return of 20 and 16 percent, respectively. There is moreover some risk associated with those investments, and the variance of the total return is

$$2x_1^2 + x_2^2 + (x_1 + x_2)^2.$$

Clearly, while investor likes large expected returns, they also like small risk. We can therefore formulate the problem that the investor wants to solve as

$$\begin{aligned}
 & \text{Max} && f(x) = 20x_1 + 16x_2 - \theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2] \\
 & \text{subject to:} && \\
 & && x_1 + x_2 \leq 5 \\
 & && x_1 \geq 0 \\
 & && x_2 \geq 0,
 \end{aligned} \tag{1}$$

where $\theta \geq 0$ is the investors' adversity to risk (so $\theta = 0$ means that the investor is risk-neutral; the larger the θ , the more risk adverse the investor is).

The model above can be generalized as follows: let μ be the vector of expected returns of n investments, Q the covariance matrix of the investments, and B a budget. Then we can write

$$\begin{aligned}
 & \text{Max} && f(x) = \mu^T x - \theta x^T Q x \\
 & \text{subject to:} && \\
 & && \sum_{i=1}^n x_i \leq B \\
 & && x \geq 0.
 \end{aligned} \tag{2}$$

Two related models are as follows. In the first, we want to maximize the expected value, while keeping the variance below a parameter $\beta \in \mathbb{R}_+$ (similarly to θ , this is defined a priori, and also a function of the risk-adversity of the investor):

$$\begin{aligned}
 & \text{Max} && f(x) = \mu^T x \\
 & \text{subject to:} && \\
 & && x^T Q x \leq \beta \\
 & && \sum_{i=1}^n x_i \leq B \\
 & && x \geq 0.
 \end{aligned} \tag{3}$$

In the second, we want to minimize the variance under a constraint guaranteeing us an expected return of at least α , again a parameter decided by the portfolio manager.

$$\begin{aligned}
 & \text{Min} && f(x) = x^T Q x \\
 & \text{subject to:} && \\
 & && \mu^T x \geq \alpha \\
 & && \sum_{i=1}^n x_i \leq B \\
 & && x \geq 0.
 \end{aligned} \tag{4}$$

Another way to measure the balance between expected returns and risk is given by the *Sharpe Ratio*

$$\frac{\mu^T x}{\sqrt{x^T Q x}}. \tag{5}$$

Clearly, we like portfolios with large Sharpe Ratio. An alternative portfolio model is therefore

$$\begin{aligned} \max \quad & g(x) = \frac{\mu^T x}{\sqrt{x^T Q x}} \\ \text{subject to:} \quad & \sum_{i=1}^n x_i \leq B \\ & x \geq 0. \end{aligned} \tag{6}$$

2.2 Least-square regression

Suppose you want to predict how the cost of an apartment in Morningside Heights depends on its size. You are given a set of pairs $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$, where:

- x_i is the size (in m^2) of apartment i ;
- y_i is the cost of apartment i .

The outcome of our prediction will be a pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, which will predict, for an apartment of size x , a price of

$$y = ax + b.$$

The *error* of the prediction on (x_i, y_i) will be

$$|ax_i + b - y_i|.$$

Our goal is to find a, b leading to “small errors”. There are many ways to formalize this idea. One of the most common is to minimize the sum of the squares of the errors. This leads to the *least-square regression* problem, defined as

$$\min_{a,b} \left\{ \sum_{i=1}^N (ax_i + b - y_i)^2 \right\}. \tag{7}$$

Note that, in the problem above, the variables are a, b , while x_i, y_i are part of the input data.

The model above can be generalized to the case we want to predict the price based on n *features* (such as size, year of construction, floor, ...). Thus, each data point (x_i, y_i) is now a $(n+1)$ -dimensional vector, with $x_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$. Our prediction will be a pair $(a, b) \in \mathbb{R}^{n+1}$ that solves

$$\min_{a,b} \left\{ \sum_{i=1}^N (ax_i + b - y_i)^2 \right\}, \tag{8}$$

where ax_i now has to be interpreted as the dot product between a and x_i . Thus, a will have one component per feature.

2.3 Sparse solutions to systems of linear equations and related problems

Suppose we are given a system of linear equations $Ax = b$ where the number of columns, n , is much larger than the number of rows, m . Standard linear algebra theory implies that, if the system has a solution, then it has infinitely many. In many applications, we are interested in a solution which has a minimum number of non-zero components. We want therefore to solve

$$\min \|x\|_0 : Ax = b, \tag{9}$$

where we define $\|x\|_0 = |\{i : x_i \neq 0\}|$. Problem (9) has a broad set of applications. Let us briefly discuss a few.

Sparse flows. Suppose we are given an undirected graph $G(V, E)$, where each node $v \in V$ has a demand/offer b_v . We want to find a flow in a directed graph obtained by arbitrarily orienting the edges of the graph, so that all demands / offers are satisfied and, among all such flows, we want one that uses a minimum number of arcs. Define a variable x_{ij} for each $ij \in E$ with $i < j$, as follows:

- $x_{ij} > 0$ if x_{ij} units of flow go from i to j ;
- $x_{ij} < 0$ if $-x_{ij}$ units of flow go from j to i .

Let $Ax = b$ be the flow constraint that require the demand/offer b_v at each node v to be satisfied. Then our goal is to solve problem (9).

Example 1. Suppose that, after a vacation together, five friends A, B, C, D, E want to send each other money so that everybody has paid the same share of the vacation costs. The amount of money each person owes / is entitled to is given below.

person	A	B	C	D	E
owes	6	4	-5	1	-6

Since they live in different countries, money can only be sent between certain pairs of friends. In particular: A can send and receive money from everyone; B can send and receive money from A, D ; C can send and receive money from A, D, E ; D can send and receive money from A, B, C, E ; E can send and receive money from A, C, D . We want to find a way A, B, C, D, E can exchange money, so that a minimum number of money exchanges is performed.

This clearly is an application of our sparse flow problem on a graph $G(V, E)$ with $V = \{A, B, C, D, E\}$ and E denoting pairs of friend that can exchange money between each other. We want in fact to solve the problem

$$\begin{aligned}
 \min \quad & \|x\|_0 \\
 & x_{AE} + x_{AB} + x_{AC} + x_{AD} = 6 \\
 & -x_{AB} + x_{BD} = 4 \\
 & -x_{AC} + x_{CD} + x_{CE} = -5 \\
 & -x_{AD} - x_{BD} - x_{CD} + x_{DE} = 1 \\
 & -x_{AE} - x_{CE} - x_{DE} = -6.
 \end{aligned} \tag{10}$$

The vectors

$$\bar{x} = \begin{pmatrix} 0 \\ 4 \\ 0 \\ 2 \\ 4 \\ -1 \\ 0 \\ 4 \end{pmatrix} \quad \text{and} \quad \tilde{x} = \begin{pmatrix} -4 \\ 5 \\ 5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 6 \end{pmatrix},$$

where components are sorted alphabetically, are feasible solutions to (10). Since $\|\tilde{x}\|_0 = 4 < 5 = \|\bar{x}\|_0$, $\tilde{x}$ is a preferred for (10).

Data compression. We have some data, collected in vector b , that we want to compress into some vector x . Matrix A encodes the “decompression” method: we can obtain b from x by computing Ax . Among all vectors x that correctly

allow us to compute b , we want one that has a smallest number of non-zero components, since these are the only components that we have to store. Thus, a solution to (9) gives us the optimal way to compress b .

Compressed sensing. There is an unknown signal $\bar{x} \in \mathbb{R}^n$ that we want to reconstruct. n is large; however, we know a priori that most components of $\bar{x}$ will be 0. We make a small set of measurements on the signal, that are collected in $Ax = b$. Among all solutions to the previous system, we want one that is sparse, since we know that our original signal was sparse. So a sparse solution is a strong candidate to be the original signal, and we can obtain such a solution by solving (9).

A problem related to (9) is that of finding sparse vectors that approximately solve the system of linear equations $Ax = b$. In formulas

$$\min \|Ax - b\|_2^2 + \gamma \|x\|_0 \quad (11)$$

where $\gamma \geq 0$ is an user-defined parameter and, for a vector y , we have $\|y\|_2 = \sqrt{\sum_{i=1}^n y_i^2}$ (ℓ_2 norm). The formulation (11) aims therefore at finding a vector that, at the same time, “almost solves” $Ax = b$ (thus, it keeps $\|Ax - b\|_2^2$ small) and is sparse (thus, it keeps $\|x\|_0$ small). The parameter γ determines how much emphasis is given to obtaining a solution that is sparse (for γ large) versus obtaining a solution that is a good approximation to a solution of the linear system $Ax = b$ (for γ small).

Let a_i be the i -th row of A . Observe that

$$\|Ax - b\|_2^2 = \sum_i (a_i x - b_i)^2.$$

Thus, (11) is equivalent to a least-square regression problem (12) plus an additional penalty on the size of the support of a :

$$\min_{a,b} \left\{ \sum_{i=1}^N (ax_i + b - y_i)^2 + \gamma \|a\|_0 \right\}. \quad (12)$$

As an application of (12), consider the problem of, at the same time, predicting the price of a house given a set of n features, and identifying which among those features are the most important in predicting the price: those are the features whose corresponding component of a is different from 0.

3 Convex sets, convex functions, and convex programs

3.1 Definitions

In order to define convex programs, we need some mathematical definitions. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called *quadratic* if all of its terms have degree 0, 1, or 2. For instance,

$$x_1^2 - x_1 x_2 + 2x_2 - 7, \quad x_1 x_2 + x_1 x_3 - 2x_2 x_3$$

are quadratic functions, while

$$x_1 x_2 x_3, \quad 2^x$$

are not.

Given a set $C \subseteq \mathbb{R}^d$, a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called *Convex* (over C) for every $0 \leq \lambda \leq 1$ and $x, y \in C$,

$$f(\lambda y + (1 - \lambda)x) \leq \lambda f(y) + (1 - \lambda)f(x). \quad (13)$$

If f is convex over $\mathbb{R}^d$, we simply say that it is convex.

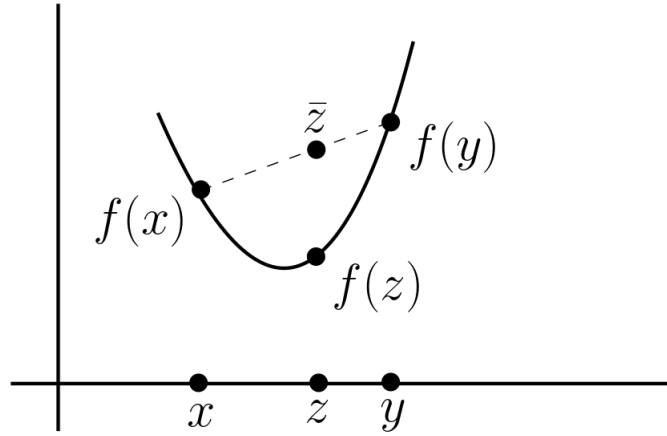


Figure 1: A convex function

To understand convexity, note that, for $0 \leq \lambda \leq 1$, the point $z = \lambda y + (1 - \lambda)x$ lies on the segment between y and x , while $\bar{z} = \lambda f(y) + (1 - \lambda)f(x)$ lies on the segment between $f(y)$ and $f(x)$. Condition 13 then implies that $f(z)$ lies below $\bar{z}$. See Figure 1 for an illustration. Note that every linear function is convex.

A set C is *convex* if $\frac{1}{2}x + \frac{1}{2}y \in C$ (i.e., C contains the middle point of x, y) for each $x, y \in C$.

Example 2. The feasible set of a linear program is convex, as well as sets of the form $\{x : \|x\|_2 \leq 1\}$ (all points contained in a ball). A set of the form $\{x : \|x\|_2 = 1\}$ is not convex since, for instance, it contains vector $(1, 0, 0, \dots, 0)$ and vector $(-1, 0, 0, \dots, 0)$, but not their middle point $(0, 0, \dots, 0)$.

We next show that $C = \{f(x) \leq \alpha\}$ is convex whenever f is convex and α is a constant. Indeed, for $x, y \in C$ and $z = \frac{1}{2}x + \frac{1}{2}y$, we have

$$\begin{aligned} f(z) &= f\left(\frac{1}{2}x + \frac{1}{2}y\right) \\ &\leq \frac{1}{2}f(x) + \frac{1}{2}f(y) \quad (\text{by convexity of } f) \\ &\leq \frac{1}{2}\alpha + \frac{1}{2}\alpha \quad (\text{since } x, y \in C) \\ &= \alpha. \end{aligned}$$

Thus, $z \in C$.

A non-linear program of the form

$$\begin{aligned} \min \quad & f(x) \\ \text{subject to:} \quad & x \in C \end{aligned} \tag{14}$$

where C is a convex set and f is a convex function over C is called a *Convex program*. If moreover f is a quadratic function, (14) is called a *Convex Quadratic program*. An illustration of the containment relations between the various classes of non-linear program is given in Figure 2.

It is important to remark that, if in (14) we replace $\min$ with $\max$, we do not obtain, in general, a convex program, since f convex does not imply, in general, that $-f$ is convex.

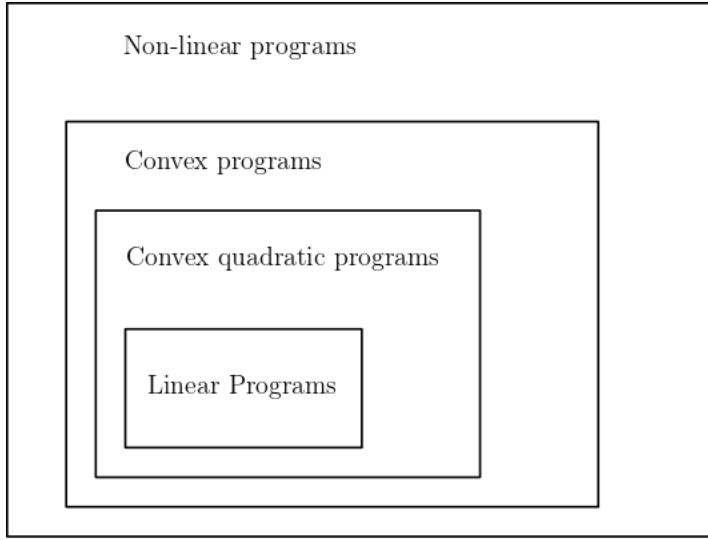


Figure 2: Convex programs and its special cases. Smaller classes usually have faster algorithms for their solutions.

3.2 Testing Convexity

In order to decide whether a non-linear program is a convex program, we need to be able to decide whether a given function is convex. Sometimes, it is possible to do so using the definition of convexity, as illustrated in the next two examples.

Example 3. A norm $\|x\|$ of a vector x is an abstraction of the classical notion of length that satisfy:

- Triangular inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Homogeneity: For $\lambda \in \mathbb{R}_{\geq 0}$, we have $\|\lambda x\| = \lambda \|x\|$.
- Positiveness: If $\|x\| = 0$, then $x = 0$.

Let us now show that $f = \|\cdot\|$ is convex, for any norm $\|\cdot\|$. We have

$$\begin{aligned}
 f(\lambda y + (1 - \lambda)x) &= \|\lambda y + (1 - \lambda)x\| \quad (\text{by definition of } f) \\
 &\leq \|\lambda y\| + \|(1 - \lambda)x\| \quad (\text{by triangular inequality}) \\
 &= \lambda \|y\| + (1 - \lambda)\|x\| \quad (\text{by homogeneity}) \\
 &= \lambda f(y) + (1 - \lambda)f(x) \quad (\text{by definition of } f).
 \end{aligned}$$

Example 4. Let $f, g : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex functions. We show that $h : \mathbb{R}^d \rightarrow \mathbb{R}$ defined as $h(x) = f(x) + g(x)$ for all $x \in \mathbb{R}^d$ is a convex function. Indeed,

$$\begin{aligned}
 h(\lambda y + (1 - \lambda)x) &= f(\lambda y + (1 - \lambda)x) + g(\lambda y + (1 - \lambda)x) \quad (\text{by definition of } h) \\
 &\leq \lambda f(y) + (1 - \lambda)f(x) + \lambda g(y) + (1 - \lambda)g(x) \quad (\text{since } f, g \text{ are convex}) \\
 &= \lambda(f(y) + g(y)) + (1 - \lambda)(f(x) + g(x)) \\
 &= \lambda h(y) + (1 - \lambda)h(x) \quad (\text{by definition of } h).
 \end{aligned}$$

It is often the case that the definition itself is not the easiest way to check if a function is convex. Fortunately, we have alternative tools, as summarized by the following lemmas, which treat the case of functions in dimension 1, 2, and n , respectively (the n -dimensional case is presented for completeness, but it will not be discussed further in this course).

Lemma 5. Let f be a 1-dimensional function that is twice differentiable. Then f is convex over C if and only if

$$f''(x) \geq 0$$

for all $x \in C$.

Lemma 6. Let f be an 2-dimensional function that is twice differentiable. Then f is convex over C if and only if

$$\frac{\partial^2}{\partial^2 x_1} f \geq 0, \quad \frac{\partial^2}{\partial^2 x_1} f \cdot \frac{\partial^2}{\partial^2 x_2} f - \left(\frac{\partial^2}{\partial x_1 \partial x_2} f \right)^2 \geq 0$$

for all $x \in C$.

Lemma 7. Let f be an n -dimensional function that is twice differentiable. Then f is convex in C if and only if all leading principal submatrices of its Hessian

$$H(f) = \begin{pmatrix} \frac{\partial^2}{\partial^2 x_1} f & \frac{\partial^2}{\partial x_1 \partial x_2} f & & \\ \frac{\partial^2}{\partial x_2 \partial x_1} f & \frac{\partial^2}{\partial^2 x_2} f & \cdots & \\ & \vdots & & \end{pmatrix}.$$

have determinant ≥ 0 .

Example 8. Let $f = x^2$. Thus, f is 1-dimensional. We compute $f''(x) = 2 \geq 0$. Thus, f is convex.

Let $f = x^3$ – again, a 1-dimensional function. We compute $f''(x) = 6x$ which is not always nonnegative. Thus, x^3 is not convex. However, it is convex over $\{x : x \geq 0\}$ since $6x \geq 0$ for $x \geq 0$.

Consider the 2-dimensional function $f(x) = \frac{1}{2}(x_1^2 + 2x_2^2)$. We compute

$$\frac{\partial^2}{\partial^2 x_1} f = 1 \geq 0, \quad \frac{\partial^2}{\partial^2 x_1} f \cdot \frac{\partial^2}{\partial^2 x_2} f - \left(\frac{\partial^2}{\partial x_1 \partial x_2} f \right)^2 = 2 - 0 \geq 0.$$

Thus, f is convex.

4 Which of the previous models is a convex program?

We now investigate which of the models from Section 2 can be formulated as convex program.

4.1 Portfolio optimization

Let us start with the portfolio optimization problem (1). To show that it can be formulated as a convex program, we first write it as a minimization problem.

$$\begin{aligned} & - \min && -f(x) = -20x_1 - 16x_2 + \theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2] \\ & \text{subject to:} && \\ & && x_1 + x_2 \leq 5 \\ & && x_1 \geq 0 \\ & && x_2 \geq 0, \end{aligned} \tag{15}$$

We then observe that the constraints form the feasible region of a linear program, hence a convex set. As for the objective function, we use Lemma 6 and compute

$$\frac{\partial^2}{\partial^2 x_1} f = 6\theta \geq 0, \quad \frac{\partial^2}{\partial^2 x_1} f \cdot \frac{\partial^2}{\partial^2 x_2} f - \left(\frac{\partial^2}{\partial x_1 \partial x_2} f \right)^2 = 24\theta^2 - 4\theta^2 = 20\theta^2 \geq 0.$$

(Recall that $\theta \geq 0$ by hypothesis). Thus (1) is a convex optimization problem. Interestingly, if the investor is risk-seeking (i.e., $\theta < 0$), the problem is not a convex problem anymore.

Now consider the other, general portfolio models. It can be shown that, for a covariance matrix Q , $x^T Q x^T$ is a convex function. This allows us to conclude that (4) is a convex optimization problem. Since every linear function is convex and the sum of convex functions is a convex function (see Example 4), we conclude that (2) is a convex optimization problem. Last, since $C = \{x : f(x) \leq \alpha\}$ is convex for f convex and a constant α (see Example 2), we deduce that (3) is convex.

Now consider the problem that maximize the Sharpe ratio (6). We are going to show that it can also be modeled as a convex optimization problem. We first introduce two new variables $z \in \mathbb{R}_+^n$, $y \in \mathbb{R}_+$, and set $z = yx$. Thus, for $y \neq 0$, we have

$$\frac{\mu^T x}{\sqrt{x^T Q x}} = \frac{\mu^T z}{y \sqrt{x^T Q x}} = \frac{\mu^T z}{\sqrt{z^T Q z}}$$

and

$$\sum_{i=1}^n x_i \leq B \Leftrightarrow \sum_{i=1}^n z_i \leq By.$$

We then rewrite (6) as

$$\begin{aligned} \max \quad & g(x) = \frac{\mu^T z}{\sqrt{z^T Q z}} \\ \text{subject to:} \quad & \sum_{i=1}^n z_i \leq By \\ & z \geq 0 \\ & y \geq 0 \end{aligned}$$

Note that z is a scaling of x (by y). Thus, we can always assume that we fix the scaling so that $\mu^T z = 1$ and rewrite the problem further as

$$\begin{aligned} \max \quad & \frac{1}{\sqrt{z^T Q z}} \\ \text{subject to:} \quad & \mu^T z = 1 \\ & \sum_{i=1}^n z_i \leq By \\ & z \geq 0 \end{aligned}$$

Now observe that maximizing $\frac{1}{\sqrt{z^T Q z}}$ is equivalent to minimizing $\sqrt{z^T Q z}$, which is equivalent to minimizing $z^T Q z$, since $\sqrt{\cdot}$ is an increasing function. Thus, (5) is equivalent to

$$\begin{aligned} \min \quad & z^T Q z \\ \text{subject to:} \quad & \mu^T z = 1 \\ & \sum_{i=1}^n z_i \leq By \\ & z \geq 0 \end{aligned}$$

which is immediately shown convex, since we observed above that $z^T Q z$ is convex when Q is a covariance matrix. Note that since the objective function of the problems from this section are quadratic or linear, all the problems are example of convex quadratic programming.

4.2 Least-square regression

The n -dimensional least square problem (12) can be shown to be convex. Let us prove it for the case $n = 2$. Since the problem has no constraint, we only need to prove that

$$\sum_{i=1}^n (ax_i + b - y_i)^2$$

is convex. Note that

$$\sum_{i=1}^n (ax_i + b - y_i)^2 = \sum_{i=1}^n (a^2 x_i^2 + b^2 + 2ax_i b + y_i^2 - 2ax_i y_i - 2by_i).$$

We employ Lemma 6 and compute (recall that the variables here are a and b , while x_i, y_i are constant)

$$\frac{\partial^2}{\partial^2 a} f = 2 \sum_{i=1}^n x_i^2 \geq 0, \quad \frac{\partial^2}{\partial^2 a} f \cdot \frac{\partial^2}{\partial^2 b} f - \left(\frac{\partial^2}{\partial a \partial b} f \right)^2 = \left(2 \sum_{i=1}^n x_i^2 \cdot \sum_{i=1}^n 2 \right) - 4 \sum_{i=1}^n x_i^2 \geq 0.$$

4.3 Sparse solutions to systems of linear equations and related problems

None of the problems (9) and (11) is convex. Indeed, $\|x\|_0$ is not convex (in particular, it is not a norm in the sense of Example 3), since we can take $x = 0, y = 1$, and set $\lambda = 1/2$ to observe

$$\left\| \frac{1}{2} \right\|_0 = 1 > \frac{1}{2} \|0\|_0 + \frac{1}{2} \|1\|_0 = \frac{1}{2}.$$

Thus, problems (9) and (11) are often solved approximately by solving instead the problems

$$\min \|x\|_1 : Ax = b \quad \text{and} \quad \min \|Ax - b\|_2^2 + \gamma \|x\|_1 \quad (\text{LASSO}),$$

which are immediately shown convex by what proved in Example 4 and Section 4.2, since $\|\cdot\|_1$ is a norm in the sense of Example 4.